

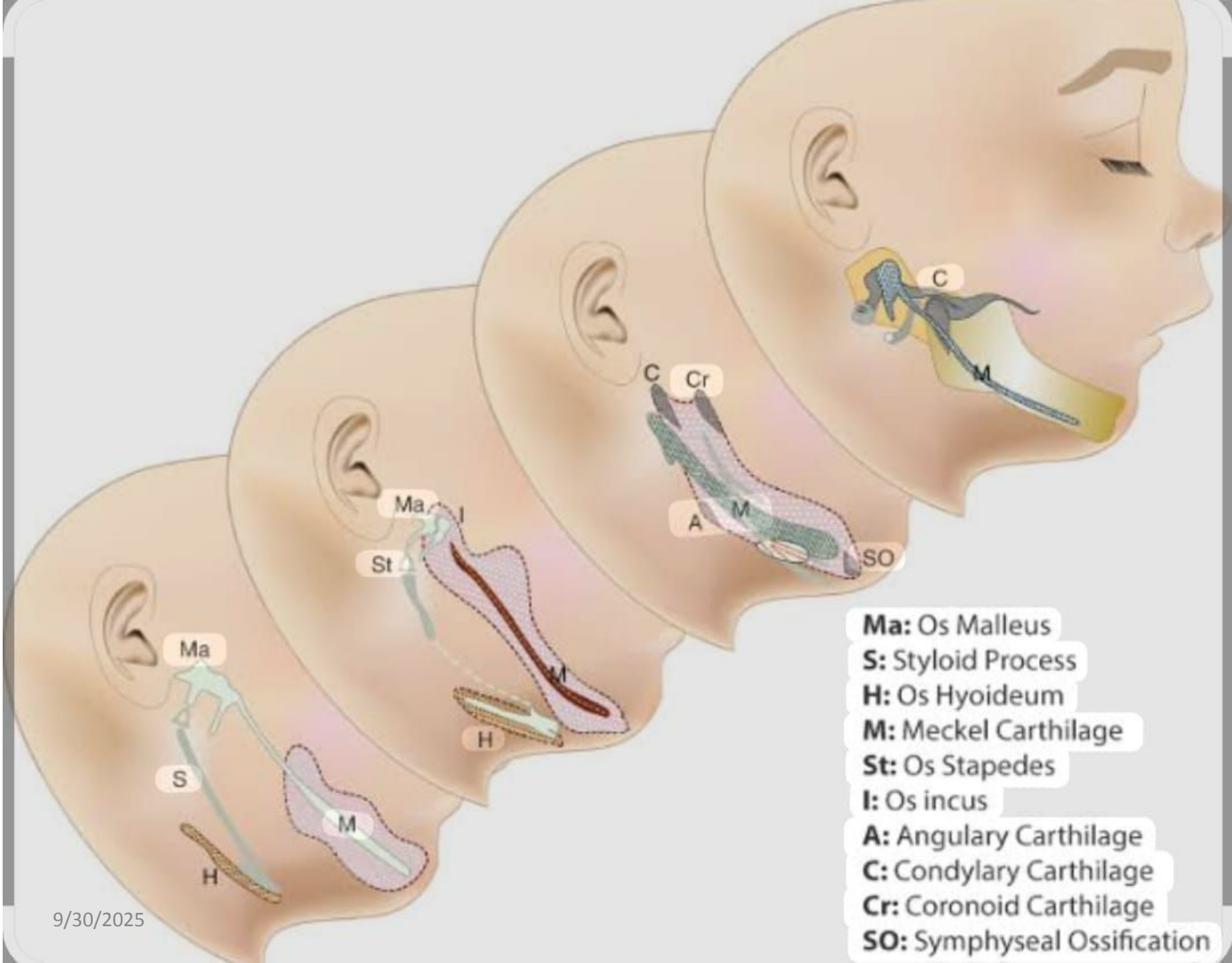
Development of Mandible, Maxilla & TMJ

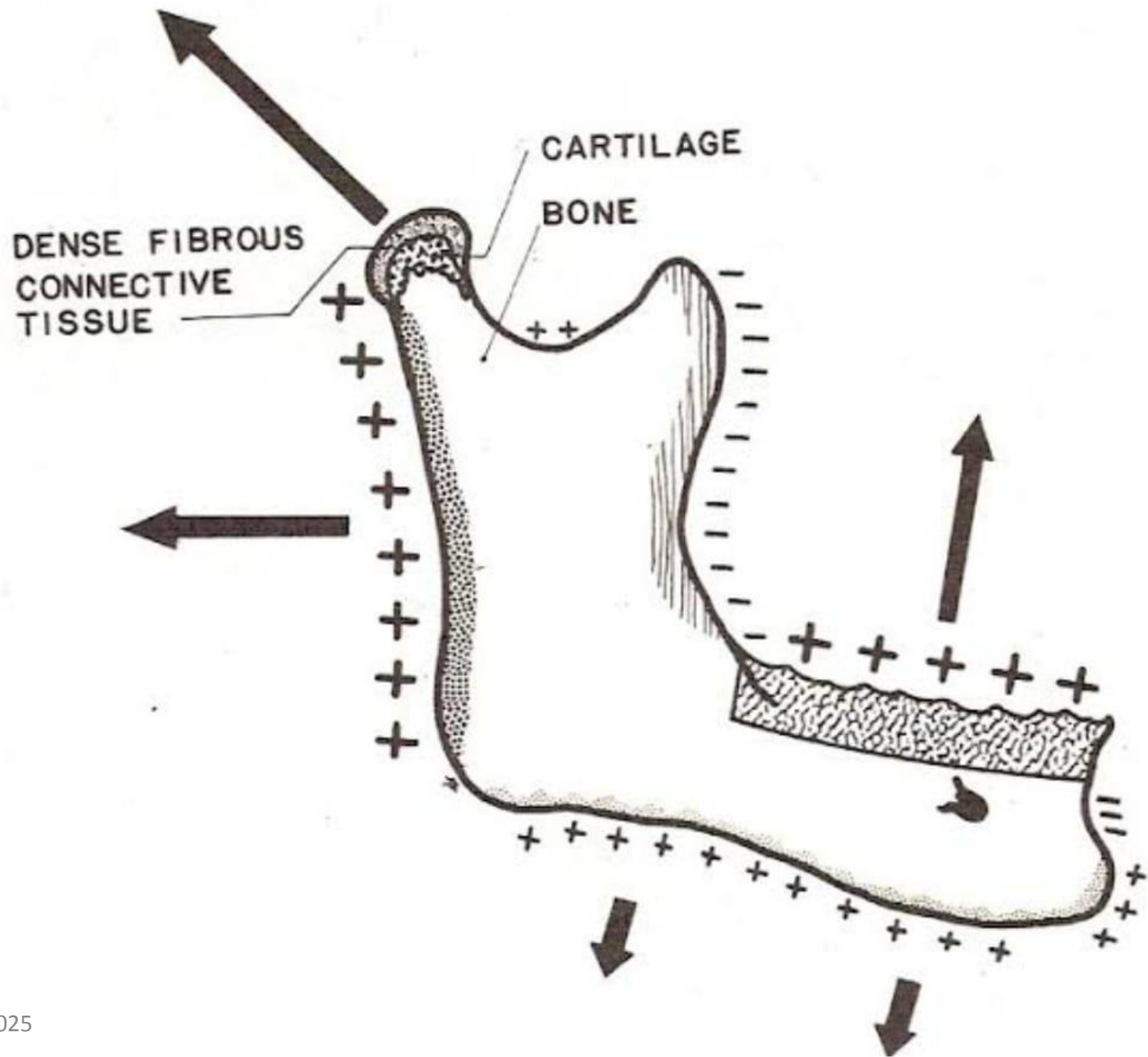
Prof.Dr.Farzeen S.Waseem

Learning outcomes

For the development of the jaws, the student should be able to:

- 1) understand the basic methods of bone growth
- 2) describe the prenatal and postnatal development of the mandible and the maxillae, highlighting the similarities and differences
- 3) appreciate the importance of modifying bone growth





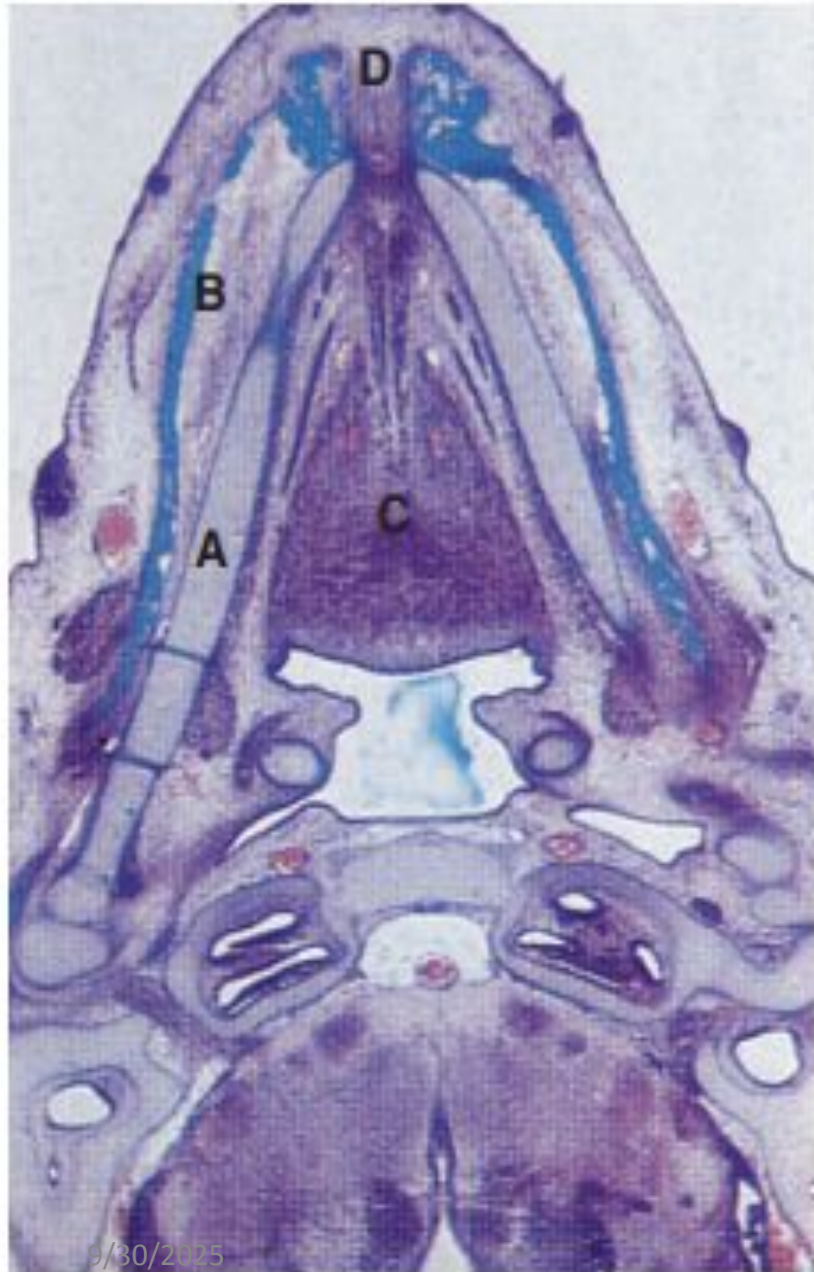
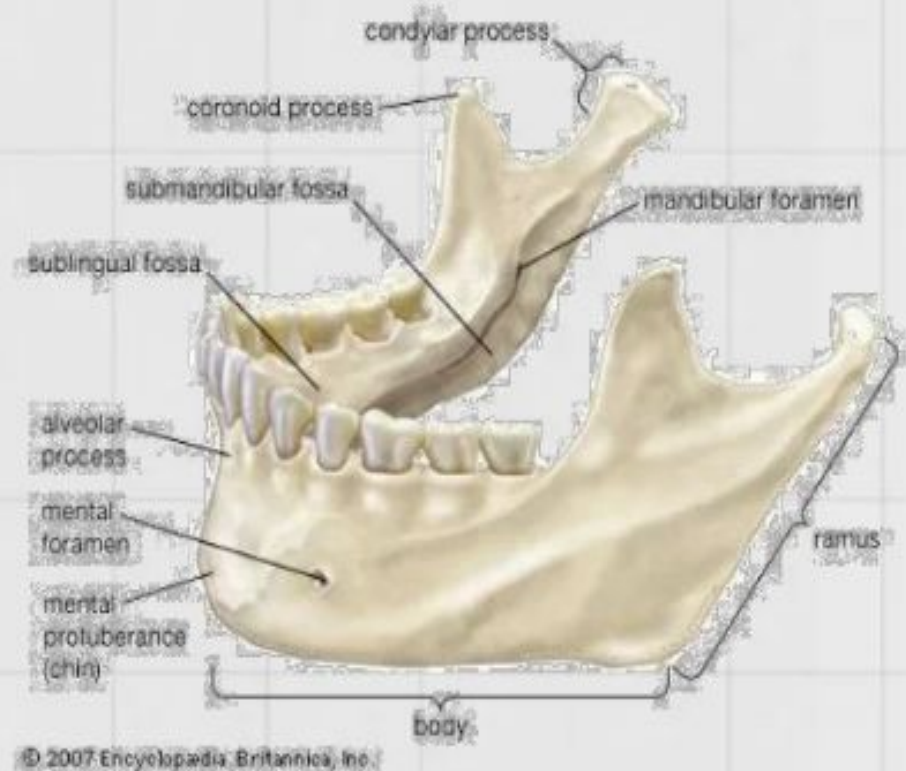


Fig. 19.1 Meckel's cartilage (A), around which the bone of the mandible (B) is forming in membrane. This is a horizontal section through the developing mandible during the 8th week of intrauterine life. Note that Meckel's cartilage extends from the cartilaginous otic capsule to the midline symphysis (D), where initially it is separated from its fellow of the opposite side by mesenchyme. C = tongue (Masson's trichrome; $\times 12$).

INTRODUCTION

- ❧ Mandible is the largest and strongest bone of the face, serves for the reception of lower teeth.
- ❧ It consists of curved, horizontal portion, the *body*; and two perpendicular portions the *rami*, which unite with the ends of the *body* nearly at right angles.



THE MANDIBLE

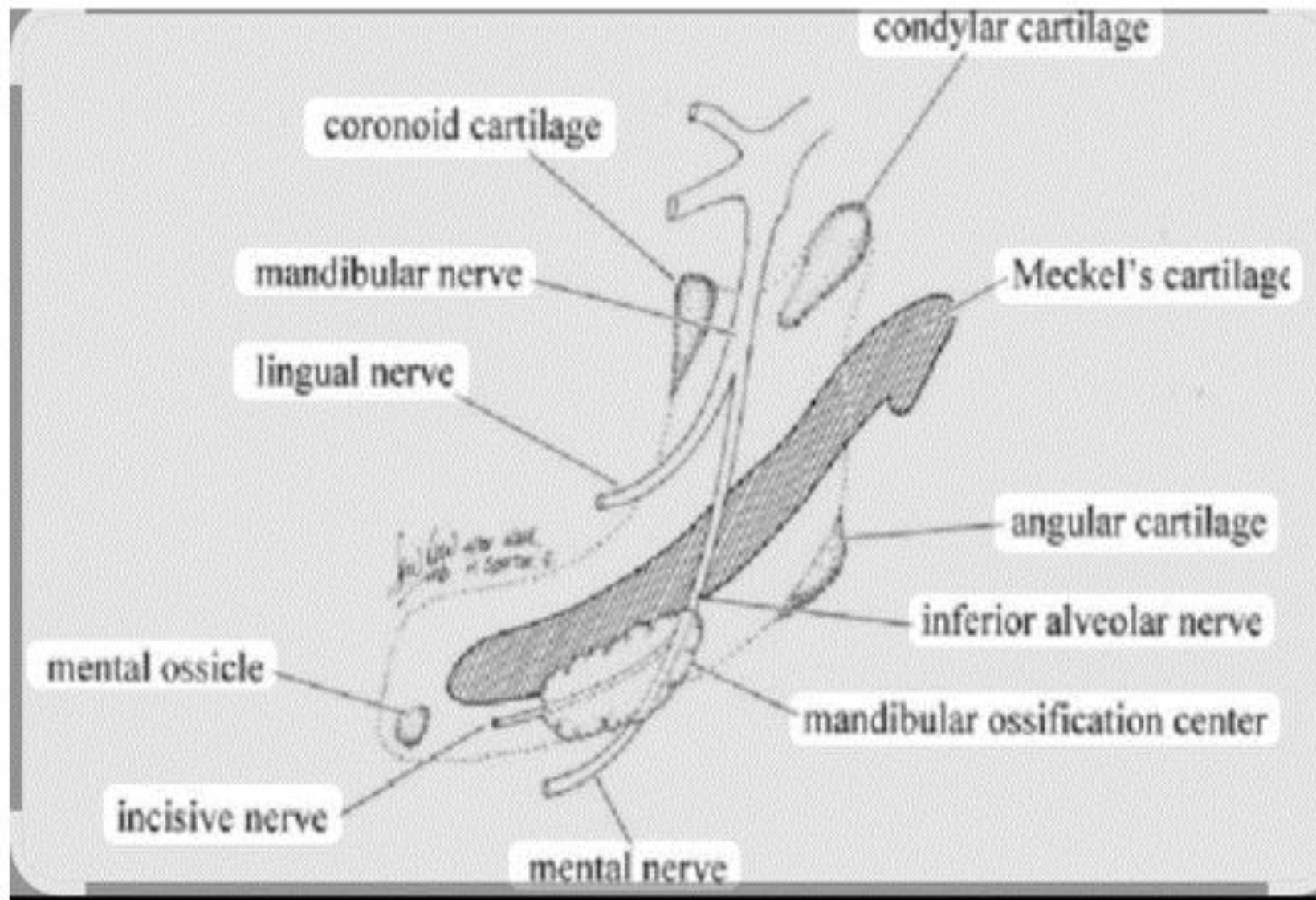
- single bone
- several skeletal units
- more soft tissue 'functional matrices' (these matrices primarily determines the growth of each skeletal unit).
- For the coronoid process forms a skeletal unit acted upon by the temporalis muscle.
- Sectioning of the temporalis muscle during early mandibular development may result in atrophy or complete absence of a coronoid process in the adult mandible.
- the alveolar process is influenced by the teeth,
- the condyle by the lateral pterygoid muscle,
- the ramus by the medial pterygoid and masseter muscles and the body by the neurovascular bundle

Development of

- ❧ Body of mandible
- ❧ Rami of mandible
- ❧ Condylar process
- ❧ Coronoid process
- ❧ Alveolar process



Courtesy:Google



GROWTH OF THE MANDIBLE

Growth of the craniofacial skeleton influence occlusal relationships and orofacial functions.

- endochondral
- intramembranous
- mandible shows both types.
- maxilla is primarily intramembranous.

THE MANDIBLE

- The mandible initially develops intramembranously.
- Appearance of secondary cartilages (the condylar cartilage being the most important).
- The developing mandible is preceded by the appearance of a rod of cartilage belonging to the first pharyngeal arch : Meckel's cartilage

THE MANDIBLE

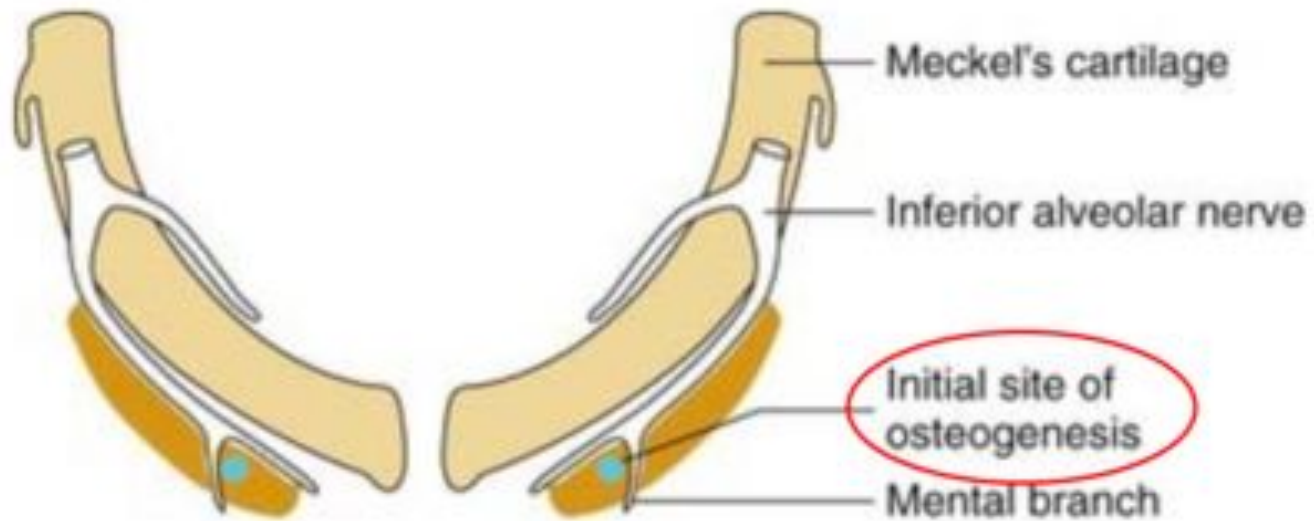
MECKEL'S CARTILAGE

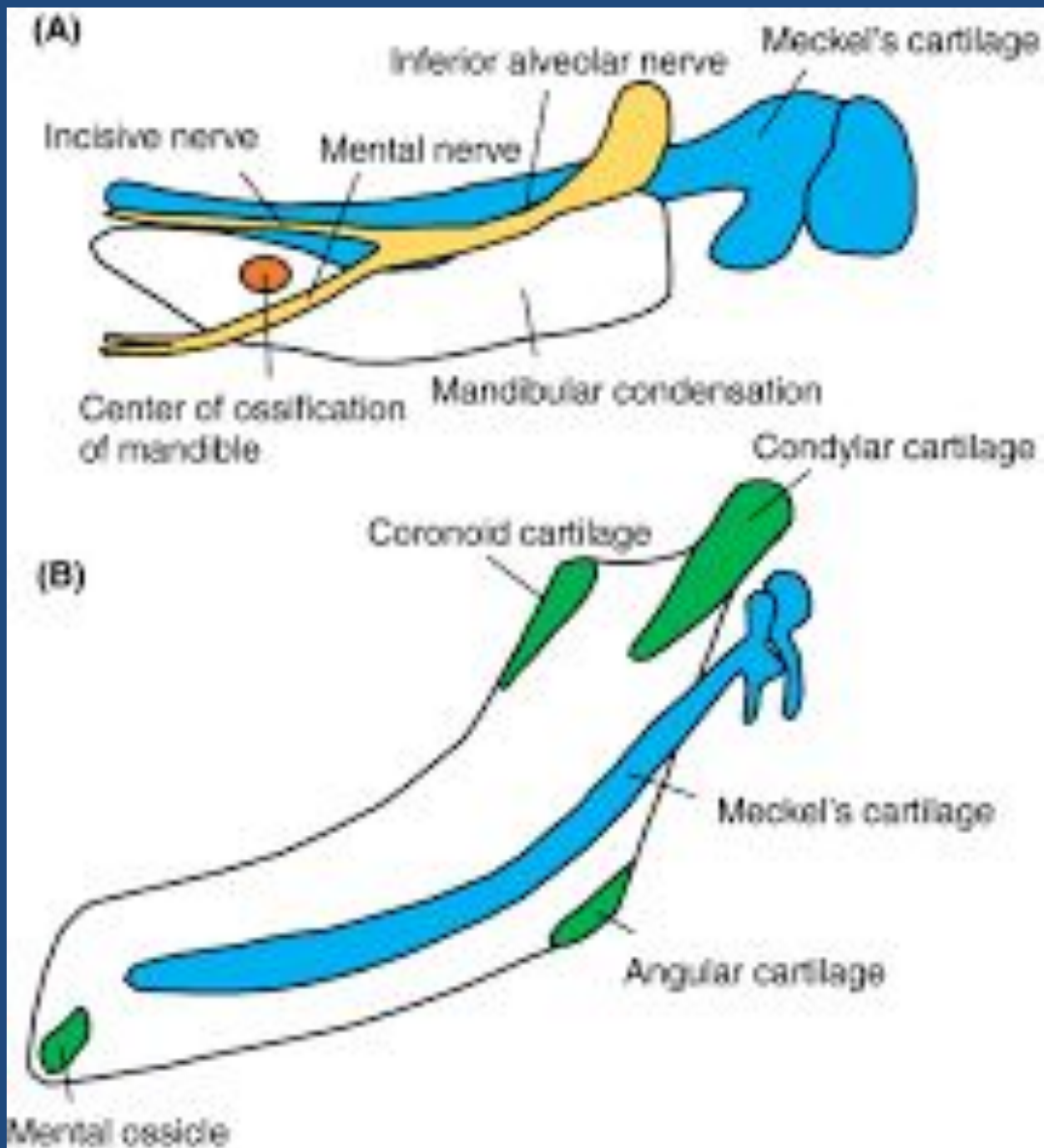
- Appears 6th week IU
- Meckel's cartilage extends from the cartilaginous otic capsule in the region of the developing ear to a midline symphysis.
- Makes little contribution to the adult mandible, merely providing a framework around which the bone of the mandible forms.
- The mandible first appears as a band of dense fibrous tissue on the anterolateral aspect of Meckel's cartilage

Intramembranous bone formation

Week

- 6th – a single ossification center close to IAV bifurcation
- 7th – bone develops lateral to Meckel's cartilage
- 8-12th – growth rate increased → mandible enlogation





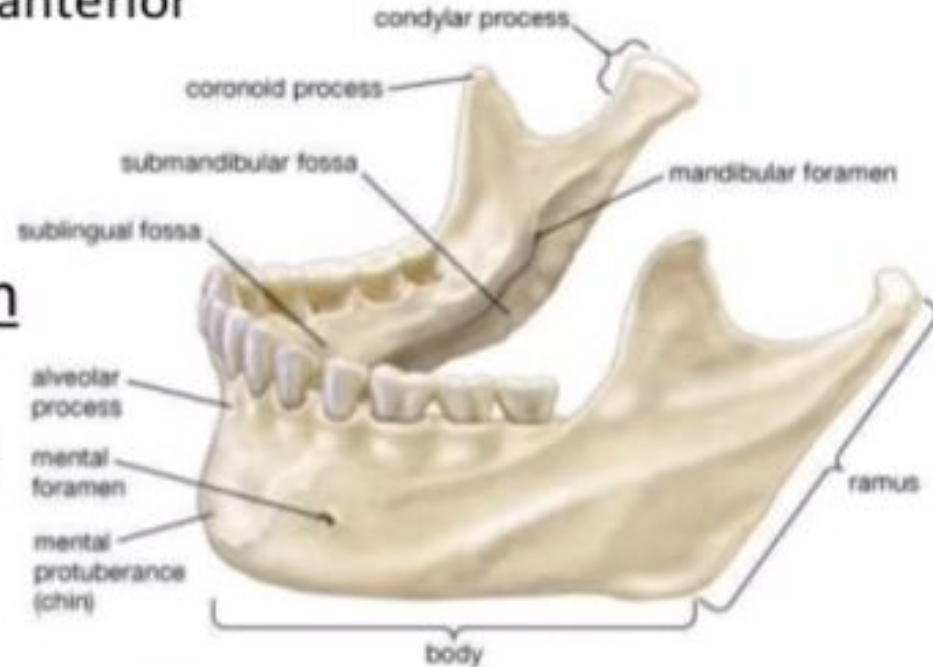
Process of ossification of mandible

- Intra membranous ossification

- Body of mandible except anterior part
- Ramus of mandible till mandibular foramen

- Endochondral ossification

- Symphysis of mandible
- Ramus above mandibular foramen
- Coronoid process
- Condylar process



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During the 7th week

- Intrauterine life, a centre of ossification appears in this fibrous tissue at a site close to the future **mental foramen**.
- From this centre, bone formation spreads rapidly backwards, forwards and upwards, and around **the inferior alveolar nerve** and its terminal branches (the incisive and mental nerves).
- Further spread of the developing bone in a forwards and backwards direction produces a plate of bone on the lateral side of Meckel's cartilage to the future body of the mandible and extends to the midline where it comes to lie in close relationship with the bone forming on the opposite side.
- The two plates of bone remain separated by fibrous tissue to form the mandibular symphysis

BODY of Mandible & Alveolar Process

- Development of the body of the mandible, continued bone formation markedly increases the size of the mandible, with
- Development of the jaws development of the alveolar process occurring to surround the developing tooth germs
- Meckel's cartilage resorbs neurovascular bundle that initially was located with the developing tooth germs now becomes contained within its own bony canal and there is considerable development of the alveolar process.
- Meckel's cartilage contributes no significant tissue to the developing mandible

Remnants of Meckel's Cartilage

- nodular remnants of cartilage
 - **the mandibular symphysis until birth**
 - **dorsal part, Meckel's cartilage ossifies to form ear ossicles (the malleus and**
 - **incus).**
- perichondrium of Meckel's cartilage persists as
 - **the sphenomandibular**
 - **sphenomalleolar ligaments.**
 - **The sphenomandibular ligament ossifies at its sites of attachment to form**
 - **the lingula of the mandible**
- As the developing tooth germs reach the bell stage (developing bone becomes closely related to it to form the alveolus).



Fig. 19.2 Transverse section through the early developing mandible (8th week of development). At this stage, only a small amount of mandibular bone has formed intramembranously on the lateral aspect of Meckel's cartilage (A). Note the beginnings of tooth development in this region as indicated by the dental lamina (B). C = tongue; D = neurovascular bundle (Masson's trichrome; $\times 60$).



Fig. 19.3 A later stage in the development of the body of the mandible. Continued bone formation has increased the size of the mandible. The alveolar process (A) grows to surround the developing tooth germ. The developing teeth share the same common crypt as the neurovascular bundle (B). Note that Meckel's cartilage (C) is now comparatively small, although it still lies medial to the developing mandibular bone. D = developing tongue (Masson's trichrome; $\times 25$).

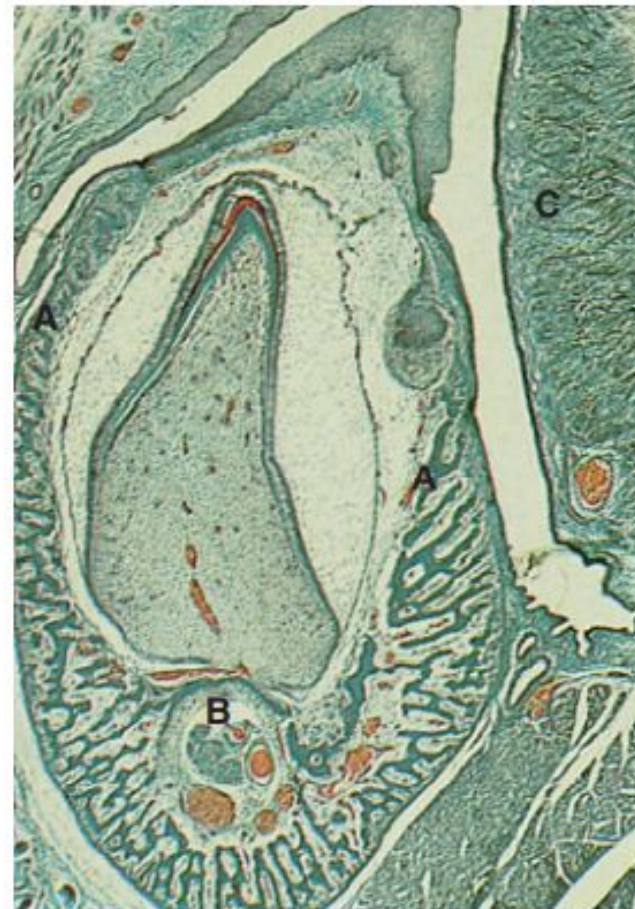


Fig. 19.4 An even later stage in the development of the body of the mandible. Meckel's cartilage has been resorbed. The neurovascular bundle (B) is now contained within its own bony canal and there has been considerable development of the alveolar process (A). C = developing tongue (Masson's trichrome; $\times 25$).

TMJ

- The TMJ develops from mesenchyme lying between the developing mandibular condyle below and the temporal bone above, which develop intramembranously.
- **During the 12th week of intrauterine life**, two clefts appear in the mesenchyme producing the upper and lower joint cavities.
- The remaining intervening mesenchyme becomes the intra-articular disc.
- The joint capsule develops from a condensation of mesenchyme surrounding the developing joint.
- At birth, the mandibular fossa is flat and there is no articular eminence; this eminence becomes prominent only following the eruption of the deciduous dentition.



Fig. 19.7 The early developing condylar cartilage (A) and temporomandibular joint. B = Meckel's cartilage; C = developing bone of mandibular fossa; D = part of developing articular disc of temporomandibular joint (decalcified section; Masson's trichrome: $\times 20$).

Growth of Mandible

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graph TD; A[Growth of Mandible] --- B[ ]; B --- C[I. Secondary cartilage]; B --- D[II. Growth of the Alveolar Process]; B --- E[III. Sub-periosteal bone formation & resorption];
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I.

Secondary cartilage

II.

**Growth of the
Alveolar Process**

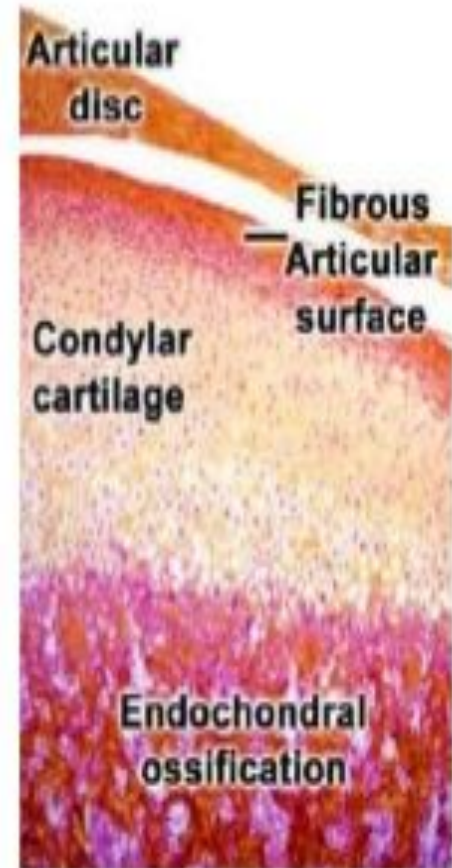
III.

**Sub-periosteal bone
formation & resorption**

Growth of Mandible

I. Secondary Cartilage

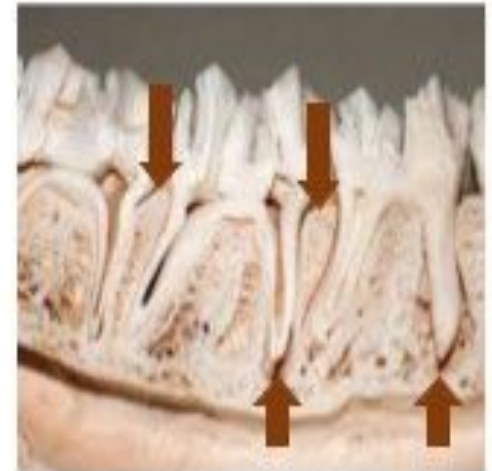
- ✓ The **most important center** of growth is the **condylar cartilage**.
- ✓ It shifts the mandible **forward** and **downward**.
- ✓ the growth of this cartilage contributes to
 1. Increase in **height** of the mandibular **ramus**
 2. Increase in the **over all length of the mandible**
 3. Increase of the **inter condylar distance**



Growth of Mandible

II. Growth of the Alveolar process

- ✓ The **distance between the 2 arches increases**; creating a space for **teeth to erupt**.
- ✓ At the same time **bone apposition** occurs at the **crest of the alveolar process** and the **fundus of the alveolus**.
- ✓ The deposited bone at the **fundus** of the alveolus counts later to the **body of the mandible**. This **increases** the distance between the **mandibular canal** and the **apices** of the premolars and first two molars.
- ✓ This means that alveolar process aids the **height of the mandible**.



Growth of Mandible

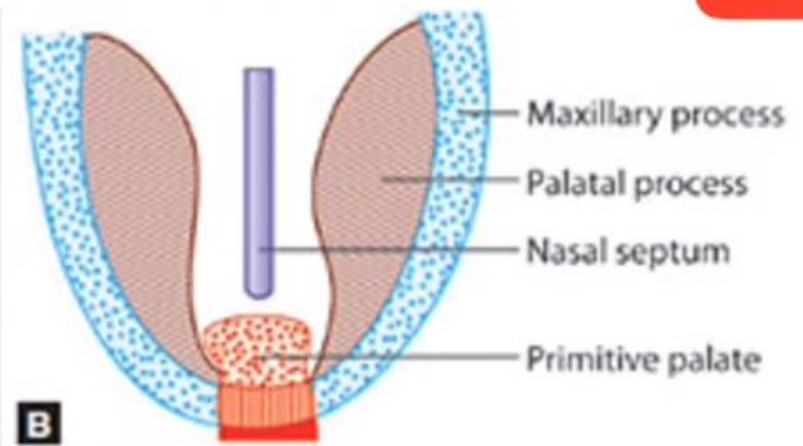
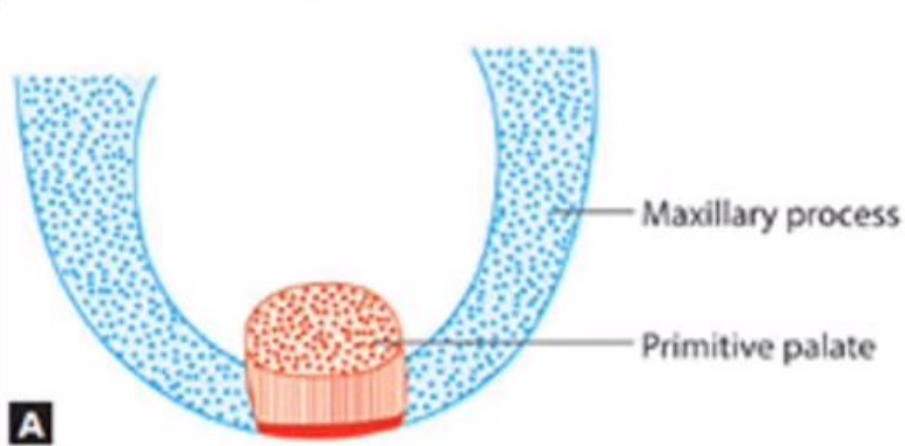
III. Sub-periosteal bone formation & resorption

The body



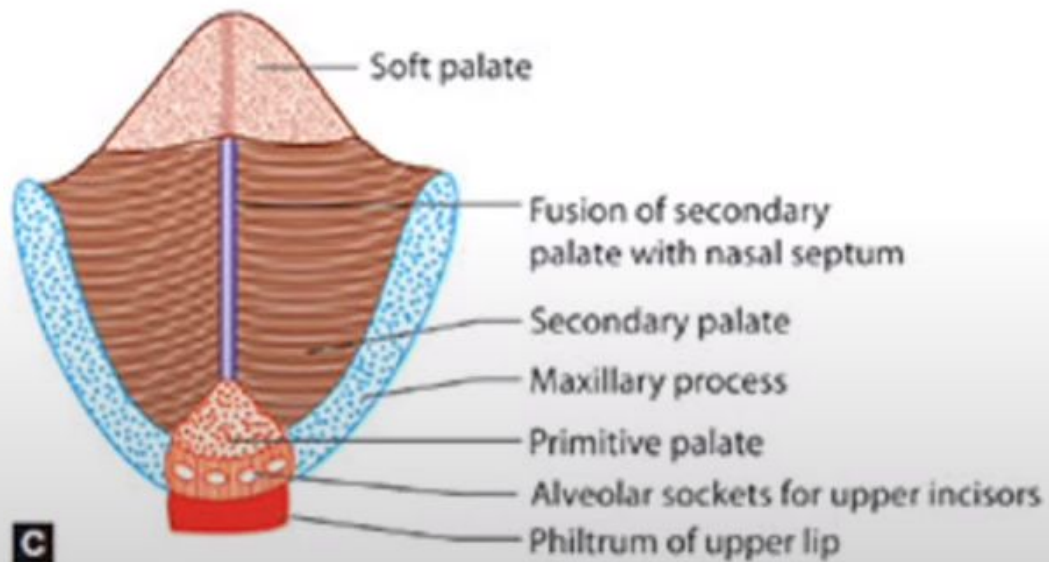
III. Subperiosteal bone apposition and bone resorption:

Bone deposition	Bone resorption	Result in
External surface of the mandible	Inner surface of the mandible	Increase the transeverse dimension
Posterior border of the remus	Anterior border of the ramus	Adjust the thickness of the ramus
Anterior border of the coronoid process	Posterior border of the coronoid process	Displacement of the coronoid process
Chin region	—————	Modeling of the lower face

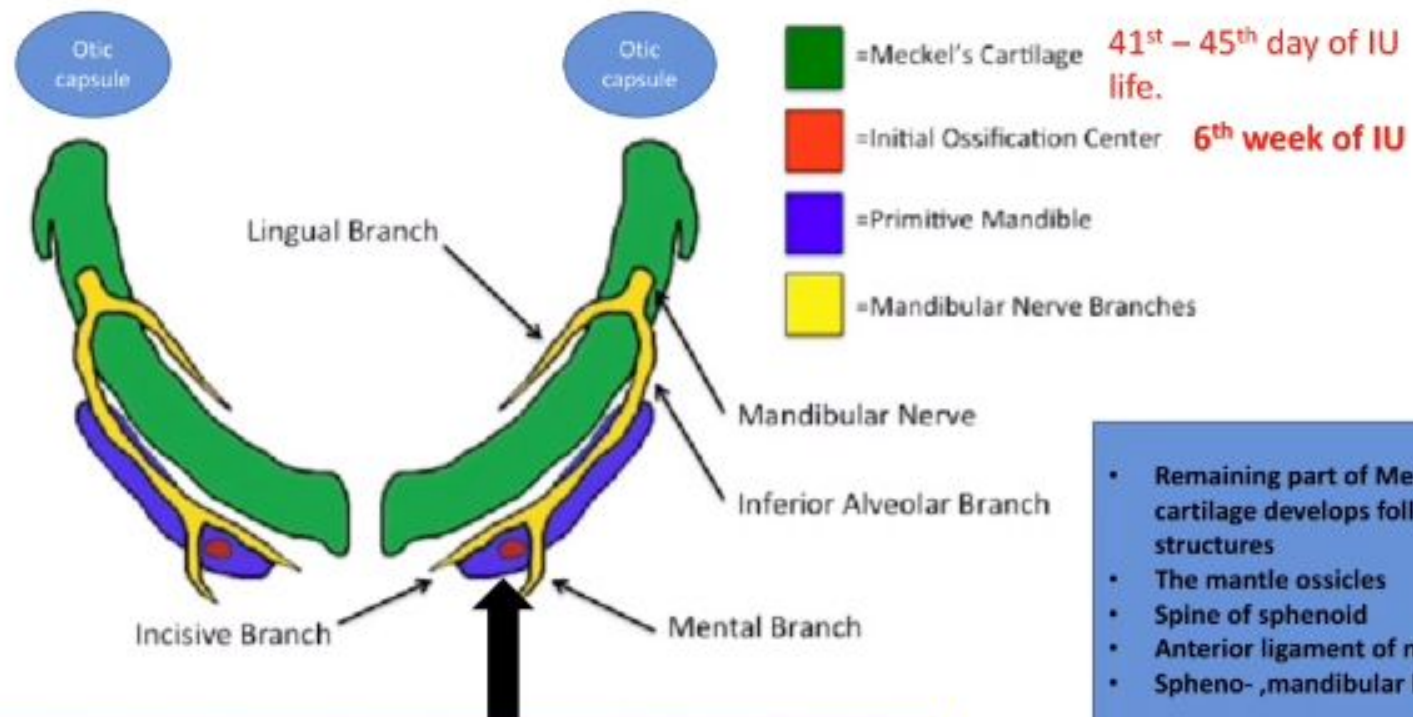


8.5 week of IU

- Intra membranous ossification
- Mid palatal suture ossifies by 12-14 years



Prenatal embryology of mandible



- Remaining part of Meckel's cartilage develops following structures
- The mantle ossicles
- Spine of sphenoid
- Anterior ligament of malleus
- Spheno- ,mandibular ligament

1st structure to develop in primordium of lower jaw - 5th cranial nerve

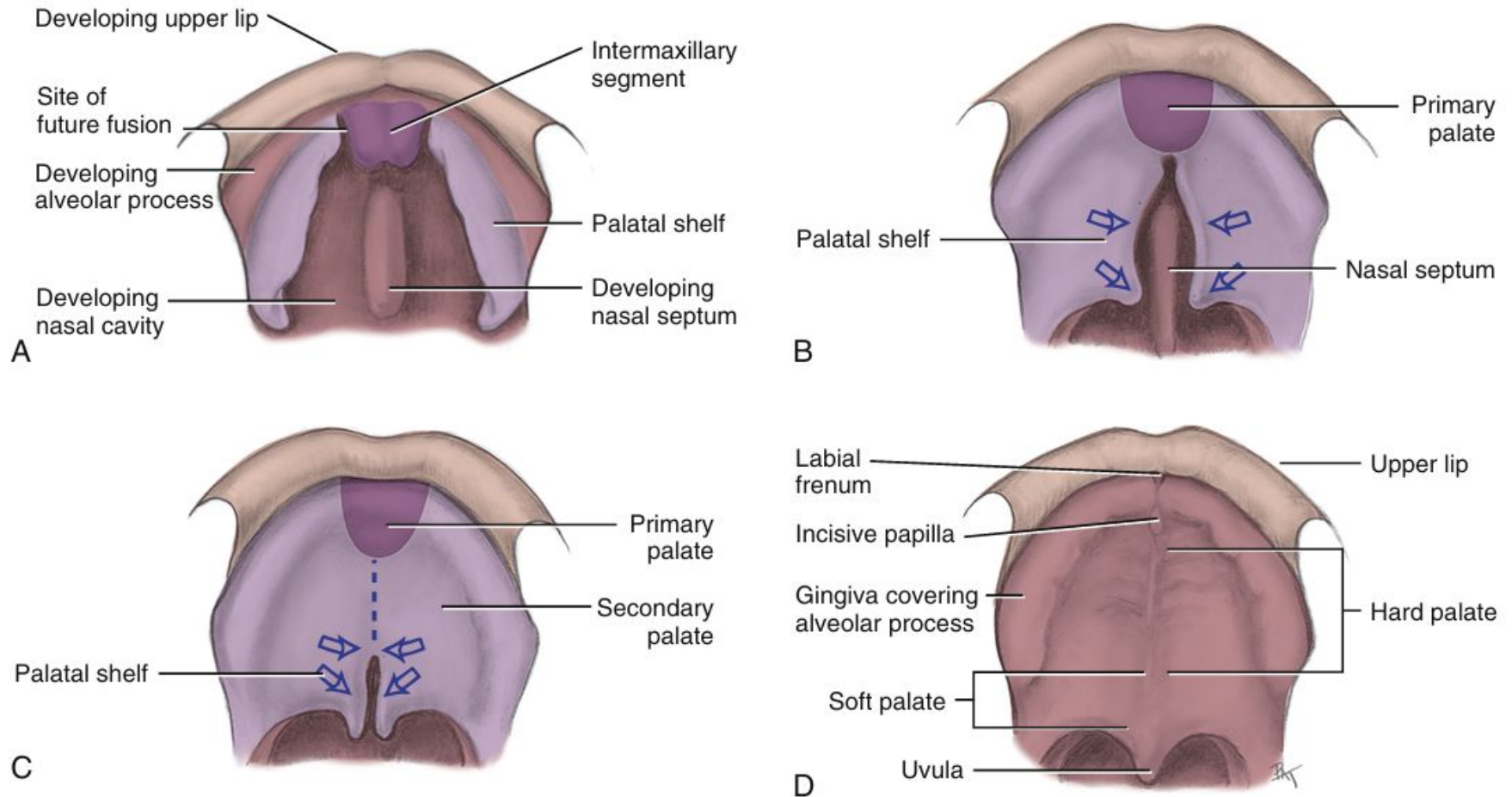


FIGURE 5-4 Development of the palate and nasal septum from an inferior view. Early stages of development of both the palate with its palatal shelves and the intermaxillary segment (**A**), formation of primary palate (**B**), nasal septum fusion with the final palate (**C**) in order to separate the nasal and oral cavities completely (**D**). Note that growth of the palatal shelves is indicated by arrows in **B** and **C** with their fusion from anterior to posterior.

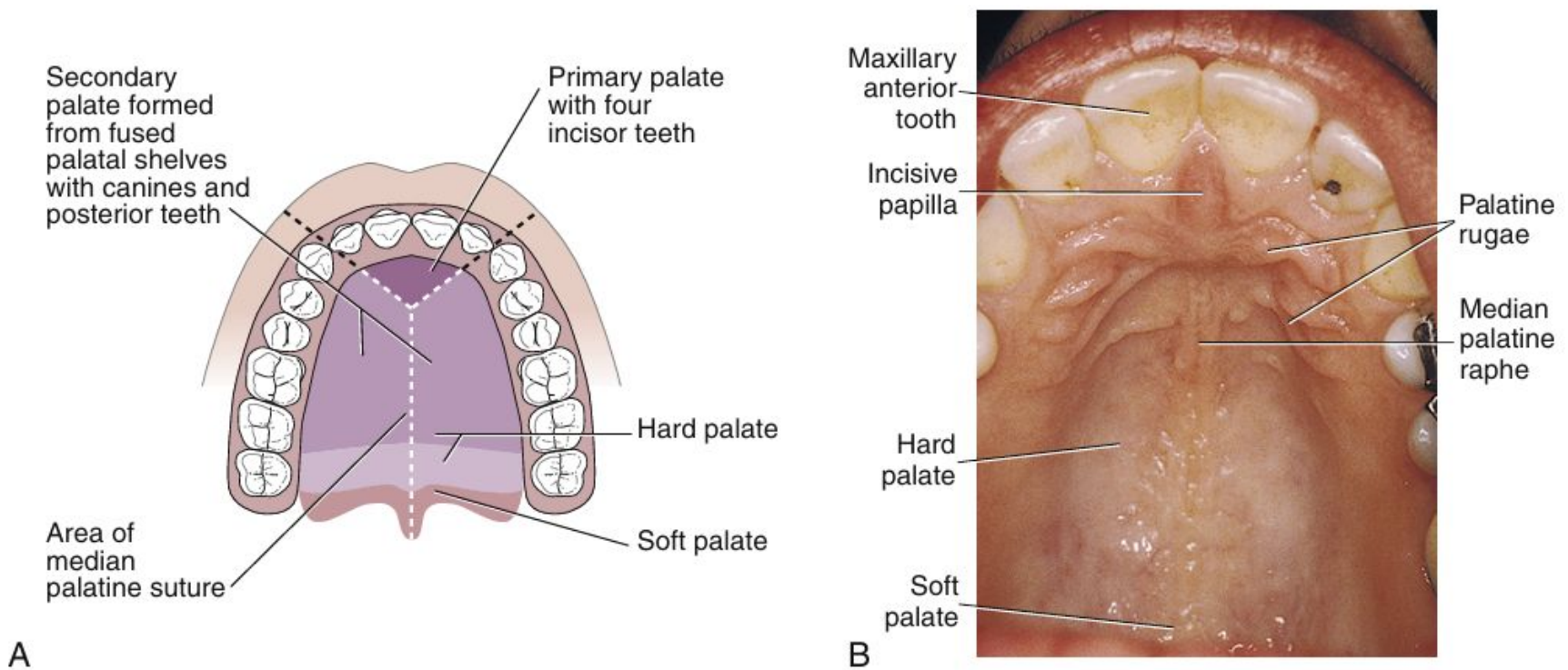
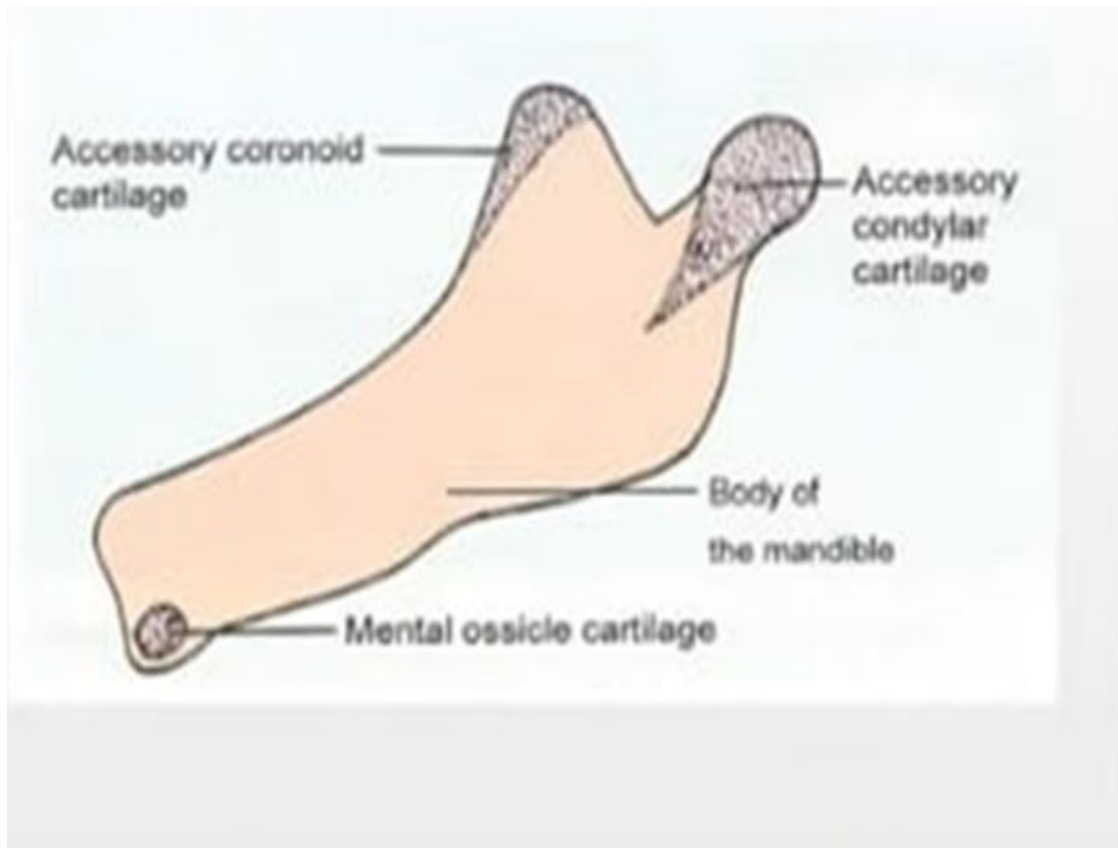


FIGURE 5-5 Adult palate with its derivative developmental structures, such as the permanent teeth. **A**, Diagram with suture areas highlighted (*dashed lines*). **B**, Mature palate with associated embryologic landmarks. (Part B courtesy of Margaret J. Fehrenbach, RDH, MS.)



ENDOCHONDRAL OSSIFICATION AREAS

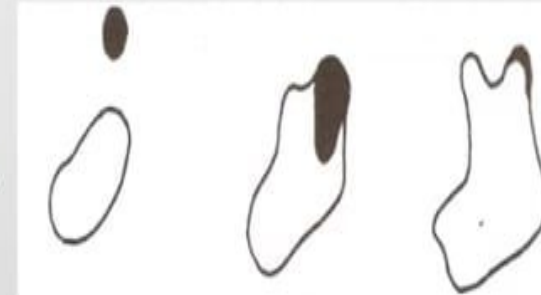


Endochondral ossification

- IT IS SEEN IN 3 AREAS OF MANDIBLE (CONDYLAR PROCESS, CORONOID PROCESS AND MENTAL REGION)

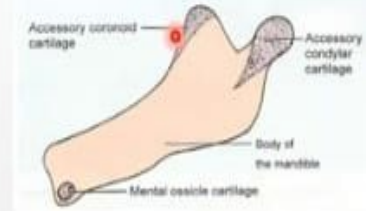
- **CONDYLAR PROCESS-**

1. AT ABOUT 5TH WEEK OF IUL, AN AREA OF MESENCHYMAL CONDENSATION CAN BE SEEN ABOVE THE VENTRAL PART OF THE DEVELOPING MANDIBLE.
2. THIS DEVELOPS INTO A CARROT/CONE- SHAPED CARTILAGE BY ABOUT 10TH WEEK AND STARTS OSSIFICATION BY 14TH WEEK.
3. IT THEN MIGRATES INFERIORLY AND FUSES WITH MANDIBULAR RAMUS BY ABOUT 4TH MONTH.
4. MUCH OF CONE SHAPED CARTILAGE IS REPLACED BY BONE BY MIDDLE OF FETAL LIFE BUT ITS UPPER END PERSIST INTO ADULthood ACTING BOTH AS GROWTH CARTILAGE AND AN ARTICULAR CARTILAGE.



• CORONOID PROCESS-

1. SECONDARY ACCESSORY CARTILAGES APPEAR IN REGION OF CORONOID PROCESS BY ABOUT 10-14 WEEK OF IUL
2. THIS SECONDARY CARTILAGE OF CORONOID PROCESS IS BELIEVED TO GROW AS A RESPONSE TO DEVELOPING TEMPORALIS MUSCLE.
3. THE CORONOID ACCESSORY CARTILAGE (TRANSIENT) BECOMES INCORPORATED INTO EXPANDING INTRAMEMBRANOUS BONE OF RAMUS AND DISAPPEAR BEFORE BIRTH.



• MENTAL REGION-

1. IN THE MENTAL REGION, ON EITHER SIDE OF SYMPHYSIS, ONE OR TWO SMALL CARTILAGES APPEAR AND OSSIFY IN 7TH MONTH OF IUL TO FORM VARIABLE NUMBERS OF MENTAL OSSICLES IN FIBROUS TISSUES OF SYMPHYSIS.
2. THESE OSSICLES BECOME INCORPORATED INTO INTRAMEMBRANOUS BONE WHEN SYMPHYSIS OSSIFIES COMPLETELY DURING 1ST YEAR OF POST-NATAL LIFE.



Fig. 1. Attachment of muscles on the lateral side

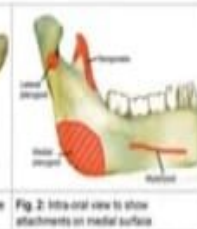
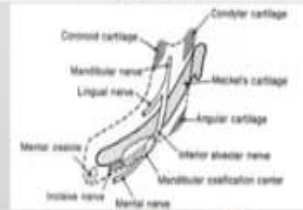


Fig. 2. Intra-oral view to show attachments on medial surface



Useful links

- <https://youtu.be/Hb1b3YPHiTE>
- <https://youtu.be/K1qdaJMyfJw>